Project Initialization and Planning Phase

Date	27 June 2026
Team ID	SWTID-2026-5245
Project Title	OptiCrop - Smart Agricultural Production Optimization Engine OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and

happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To build an intelligent crop recommendation system that leverages machine learning to analyze soil and climate parameters (N, P, K, temperature, humidity, pH, rainfall) and recommend the optimal crop, maximizing yield and resource efficiency for farmers.
Scope	The project covers data collection and preprocessing, training a classification model on historical crop-environment data, and deploying a web-based tool that provides real-time crop recommendations, crop suitability assessments, and analytics for farmers, researchers, and policymakers.
Problem Statement	
Description	Farmers often lack access to a reliable, data-driven way to determine which crop best matches their specific soil nutrients and climate conditions, leading to poor crop selection, reduced yields, and wasted resources
Impact	Solving this will help farmers make informed, data-backed planting decisions, improve crop yields and resource efficiency, and give researchers and policymakers better insight into crop-environment relationships for sustainable agricultural strategies
Proposed Solution	

Approach	Employing a supervised machine learning classification model trained on soil (N, P, K, pH) and climate (temperature, humidity, rainfall) data to predict the most suitable crop, exposed to users through a simple, web-based interface
Key Features	- ML-based crop recommendation engine trained on soil and climate data. - Crop suitability / environmental assessment for a chosen crop. - Resource-efficiency tips alongside the recommended crop. - Simple web-based interface accessible to farmers with minimal technical expertise. - Analytics view for researchers and policymakers to study crop-environment patterns.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (dual/quad-core); no GPU required – model is a lightweight classifier (e.g., Random Forest / Decision Tree)
Memory	RAM specifications	4–8 GB
Storage	Disk space for data, models, and logs	< 1 GB (CSV dataset + serialized model files)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE	Jupyter Notebook / VS Code; deployed on Vercel
Data		
Data	Source, size, format	Kaggle Crop Recommendation dataset, ~2200 rows, csv (N, P, K, temperature, humidity, pH, rainfall, label)